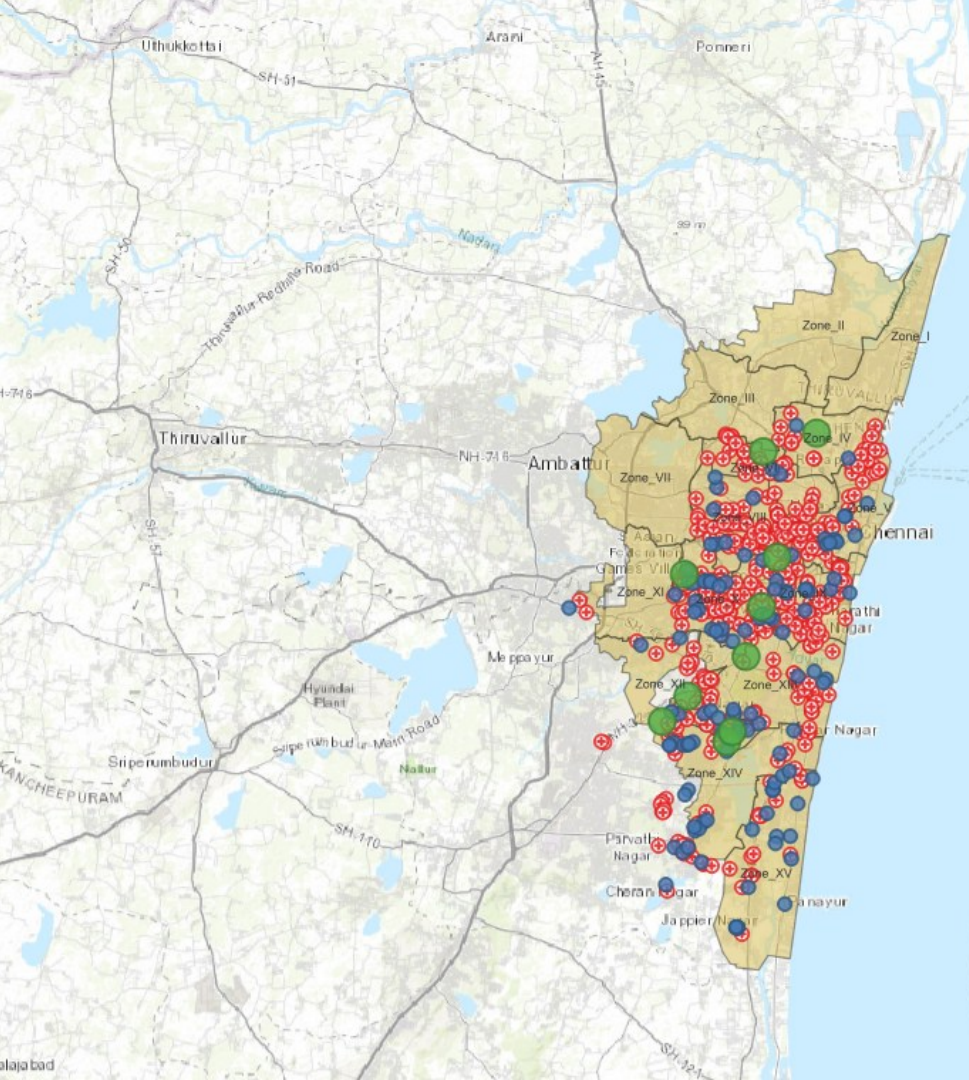




Assessing Public Service Accessibility Using GIS & Python

**Greater Chennai Corporation,
Tamil Nadu**

Kushi Kondoju

Introduction

Urban accessibility to public services (healthcare, education, recreation) is a key indicator of equitable urban development. In rapidly growing cities like Chennai, uneven distribution of services and transport constraints can lead to accessibility gaps. This study uses GIS and network-based analysis to evaluate spatial accessibility and identify underserved areas.

Aim
To evaluate the **spatial accessibility of public services** in Chennai using an integrated **GIS and Python-based analytical framework**, in order to identify underserved areas

Objectives

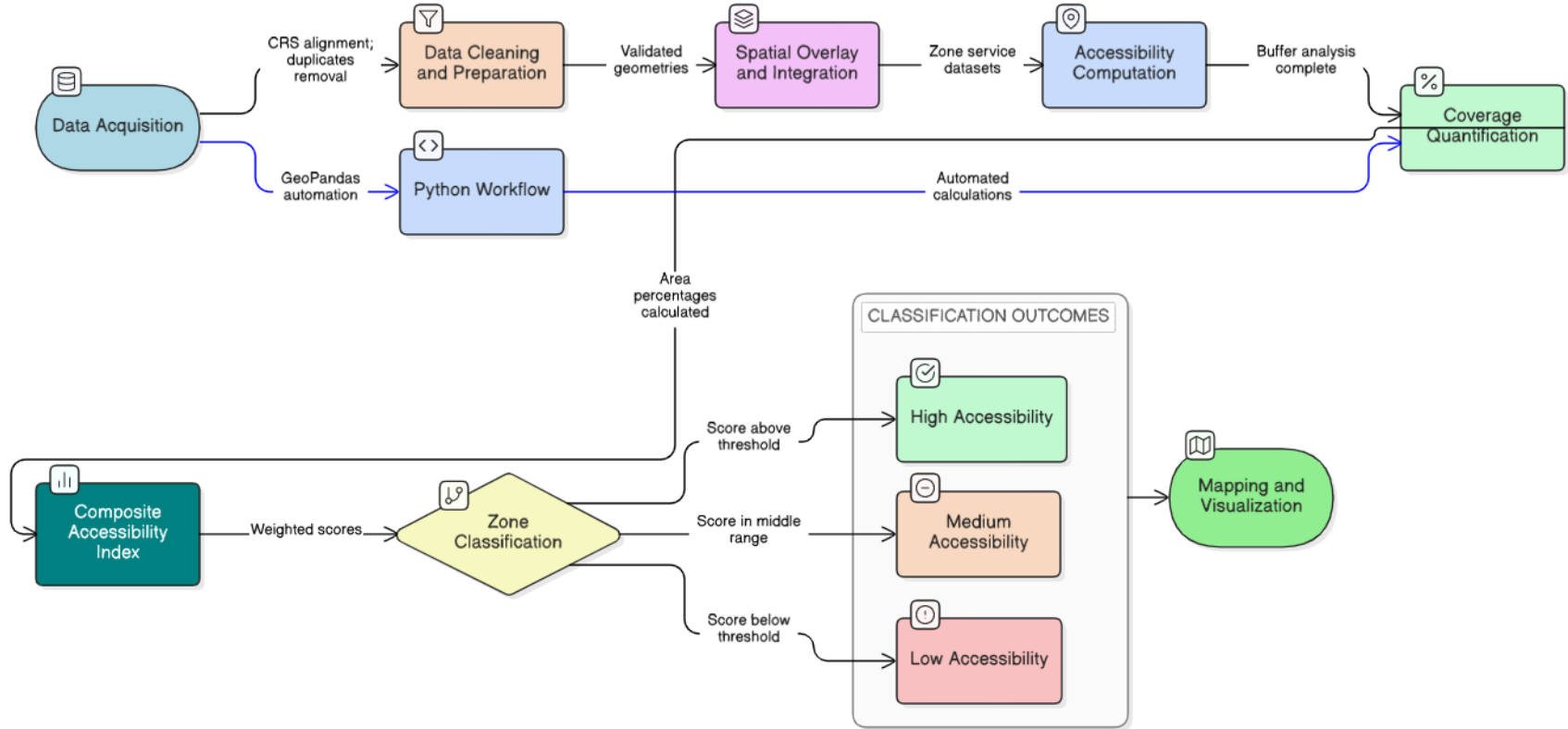
- To collect and preprocess spatial datasets for public services (healthcare, education, parks)
- To analyse spatial distribution and service coverage across different zones
- To compute accessibility using proximity-based and spatial analysis techniques
- To identify areas with low accessibility and service gaps
- To develop a reproducible workflow using Python for scalability and automation
- To generate maps and insights that support planning decisions

Rapid urbanization in Chennai has resulted in:

- Uneven service distribution
- Transport-related accessibility constraints
- Traditional planning lacks data-driven accessibility evaluation



Methodology



Study Area

Study Area: Greater Chennai Corporation (GCC), Tamil Nadu

Greater Chennai Corporation (GCC), located along the Bay of Bengal, is a major metropolitan region and an important economic and administrative hub of Tamil Nadu. The city is divided into multiple zones with varying levels of density, infrastructure, and service distribution, ranging from well-developed central areas to rapidly growing peripheral zones.

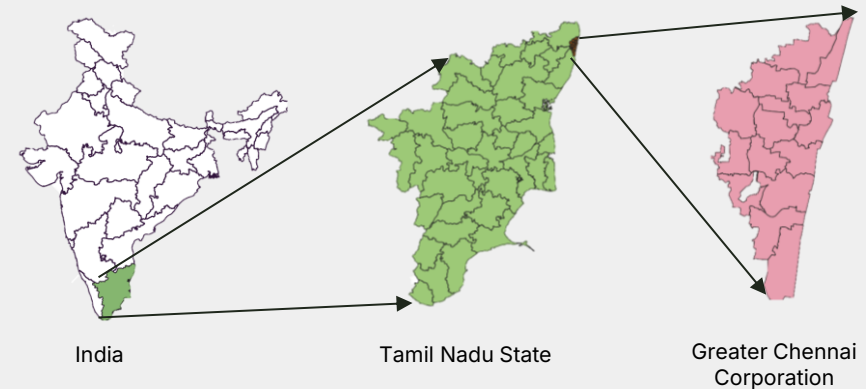
Rapid urbanization has led to uneven distribution of public services and differences in accessibility across zones, making Chennai an ideal case for evaluating urban service accessibility using geospatial techniques.

Datasets & Sources

- **Administrative Boundaries (GCC Zones)**
 - Source: Government / open GIS data portals
- **Healthcare Facilities (Hospitals) , Educational Institutions (Schools), Recreational Spaces (Parks)**
 - Source: OpenStreetMap (OSM) via Overpass Turbo

Tools & Python Libraries Used

- **GeoPandas** – Spatial data handling and analysis
- **Pandas** – Data processing and calculations
- **Shapely** – Geometry operations (buffer, intersection)
- **Matplotlib** – Static map visualization
- **Leafmap / Folium** – Interactive mapping
- **NumPy** – Numerical computations
- **PyProj** – Coordinate system transformations

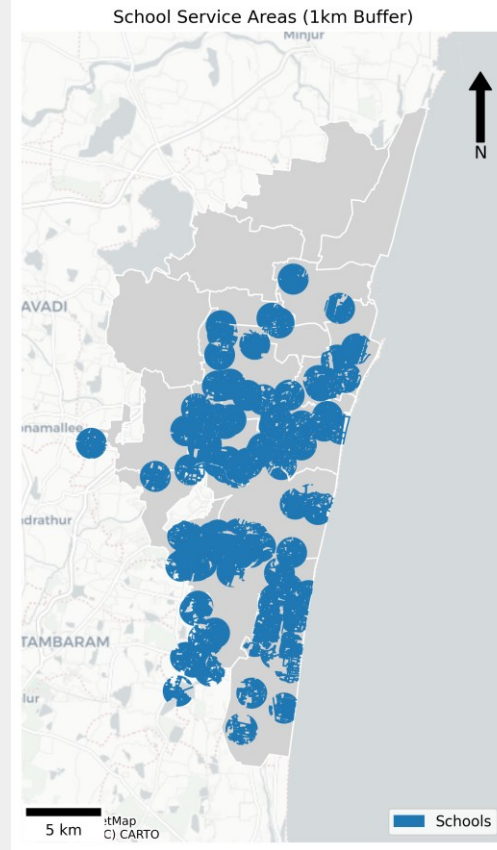


Network-Constrained Service Coverage Analysis

This analysis evaluates public service accessibility using a two-stage buffer approach that incorporates both proximity and network connectivity. Buffers are first generated around service locations (schools, hospitals, and parks) to represent their theoretical coverage.

To improve accuracy, a **50-meter buffer is created around the road network**, and service buffers are intersected with it to ensure that only areas connected through roads are considered accessible.

This method provides a more realistic representation of accessibility by accounting for actual movement pathways rather than simple **Euclidean distance**.



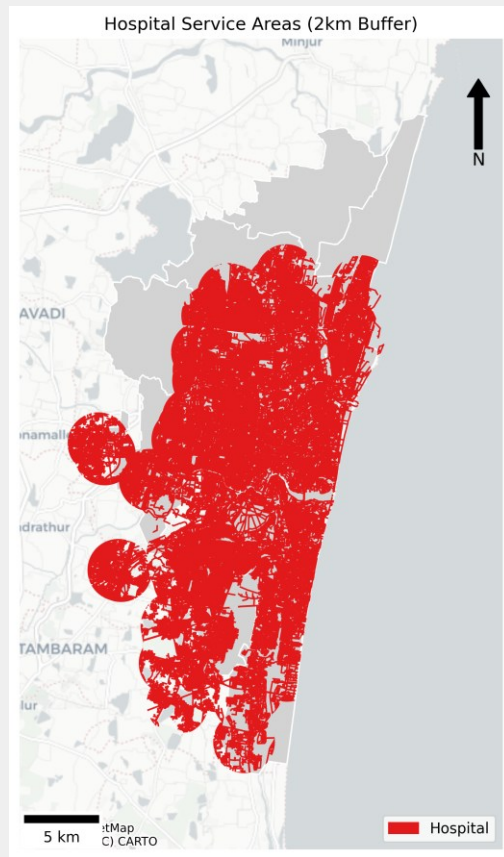
Spatial Operations (Python Workflow)

- `buffer()` → Create 1 km service area around schools
- `buffer()` → Create 50 m road network buffer
- `intersection()` → Constrain service area to road access
- `plot()` → Visualize using Matplotlib

Inference

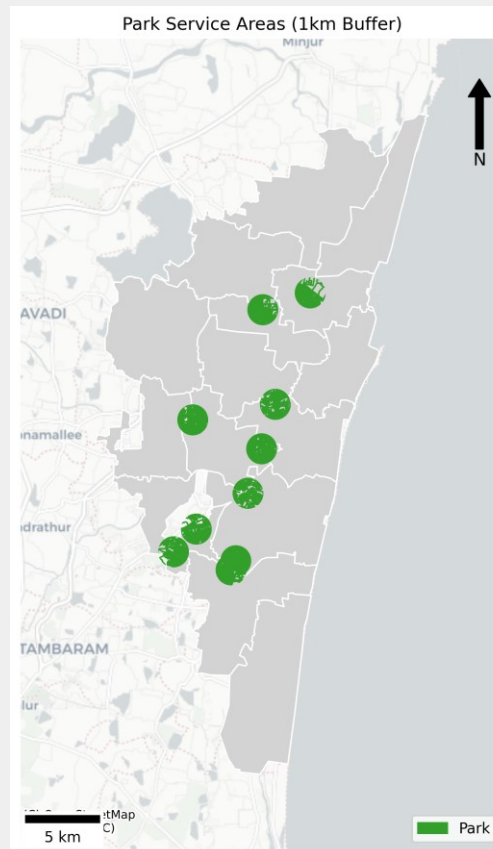
- High concentration in central zones
- Strong coverage along road networks
- Peripheral areas show gaps
- Coverage depends on road connectivity

Network-Constrained Service Coverage Analysis



Inference

- Hospitals show wide spatial coverage due to larger buffer distance (2 km)
- Coverage is dense and continuous in central zones
- Strong accessibility observed along major road networks
- Peripheral areas still show comparatively lower reach



Inference

- Parks exhibit limited and scattered distribution
- Coverage is low and discontinuous across most zones
- Accessibility is mainly clustered in select areas
- Many zones show minimal or no park access

Limitation (Parks)

- Park dataset is limited and may not capture all existing spaces
- Analysis is included mainly to demonstrate the workflow, not exact real-world availability

Zone-wise Accessibility Analysis (%)

Accessibility is measured as area-based coverage within each zone

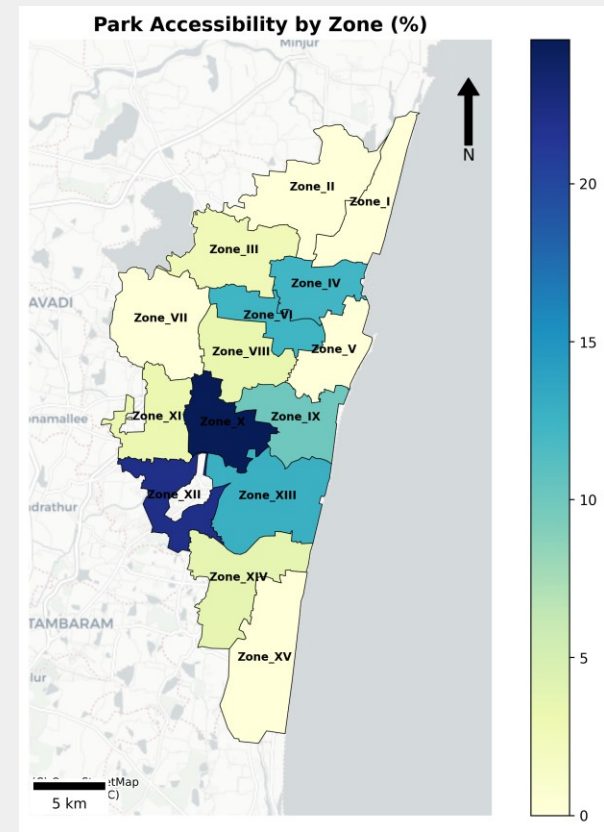
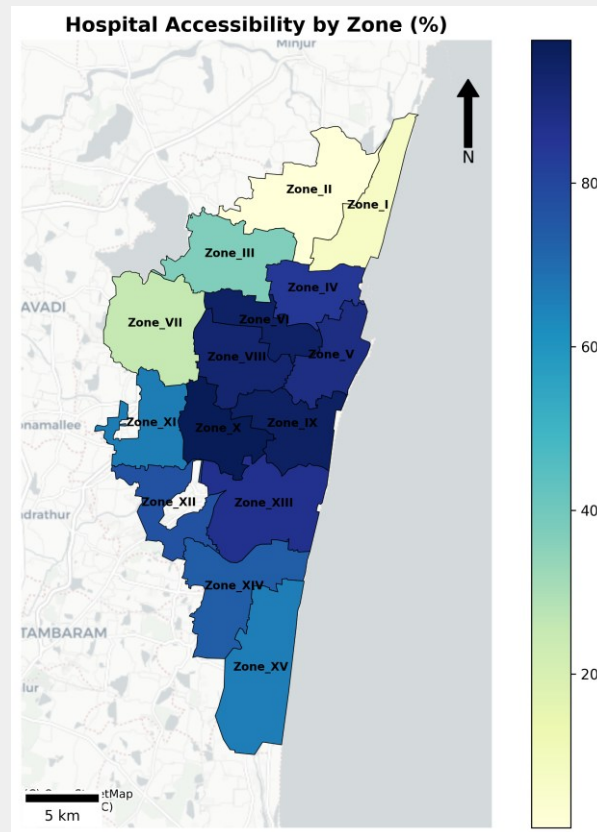
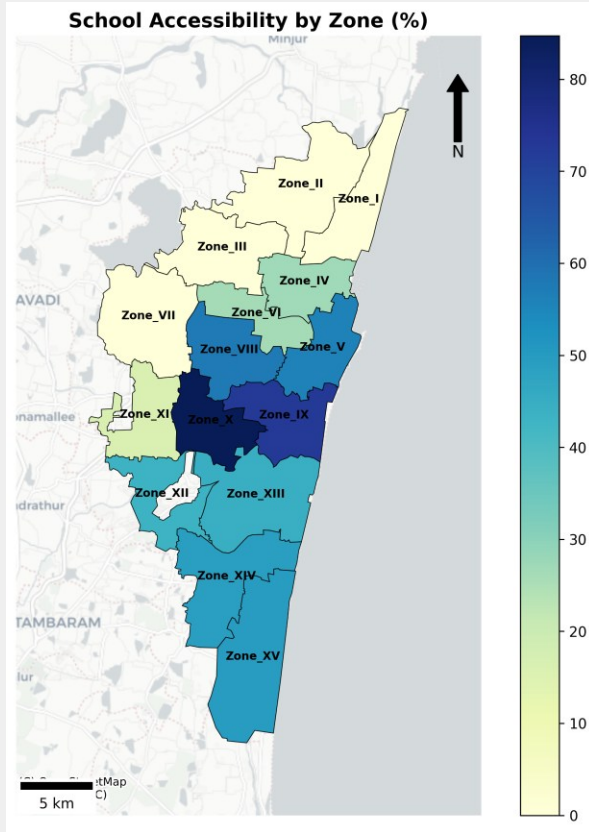
For each service:

$\text{Accessibility (\%)} = (\text{Service Covered Area} / \text{Total Zone Area}) \times 100$

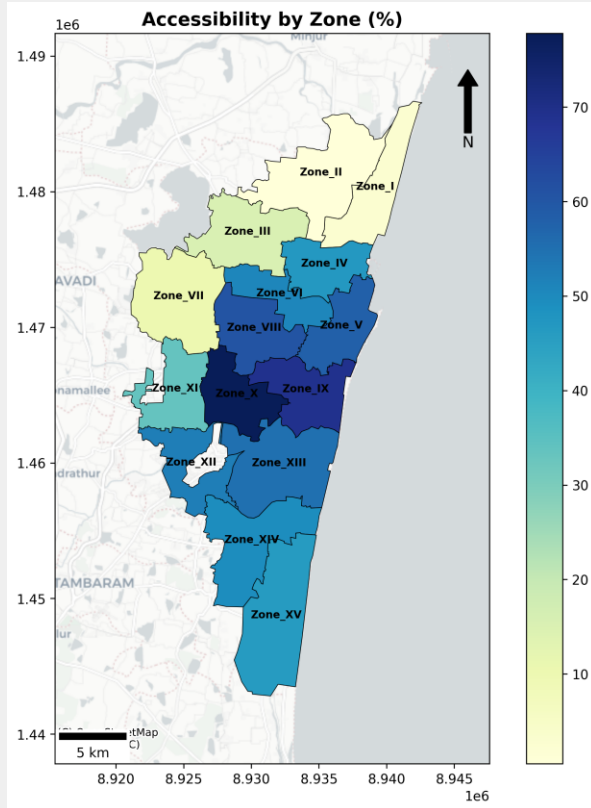
Python Workflow

- `overlay()` → Intersect service coverage with zones
 - `area` → Calculate covered and total zone area
 - `groupby()` → Aggregate zone-wise values
 - `calculate %` → Compute accessibility score
 - `merge()` → Join results back to zones
 - `plot()` → Visualize thematic maps
- The analysis highlights clear variations in service distribution across Greater Chennai, with hospital accessibility consistently high due to larger buffer distance and better spatial distribution, resulting in wider coverage across most zones.
 - School accessibility shows a balanced pattern, with moderate to high coverage in central and southern zones, while some peripheral areas experience comparatively lower accessibility levels.
 - In contrast, park accessibility is significantly lower and uneven, indicating a lack of recreational spaces in many areas, especially in outer zones; overall, central zones have better access to services, whereas peripheral zones show noticeable gaps.

Zone-wise Accessibility Analysis (%)



Combined Accessibility Score & Zone Classification



Zone No	Zone Name	Accessibility Score	Access Level
X	Kodambakkam	1.000	High
IX	Teynampet	0.888	High
VIII	Anna Nagar	0.779	High
V	Royapuram	0.743	High
XIII	Adyar	0.707	High
XII	Alandur	0.670	Medium
VI	Thiru-Vi-Ka Nagar	0.651	Medium
XIV	Perungudi	0.633	Medium
IV	Tondiarpet	0.599	Medium
XV	Sozhanganallur	0.592	Medium
XI	Valasaravakkam	0.427	Low
III	Madhavaram	0.193	Low
VII	Ambattur	0.125	Low
I	Thiruvotriyur	0.029	Low
II	Manali	0.000	Low

Accessibility scores are derived by combining the individual accessibility values of schools, hospitals, and parks into a single composite index.

A **weighted approach** is used to reflect the relative importance of each service:

Schools → 40%

Hospitals → 40%

Parks → 20%

Combined Accessibility Score =
 $(0.4 \times \text{School Accessibility}) + (0.4 \times \text{Hospital Accessibility}) + (0.2 \times \text{Park Accessibility})$

Access Level

High → > 0.7

Medium → 0.4 – 0.7

Low → < 0.4

Key Findings & Planning Implications

KEY FINDINGS

- Accessibility is **unevenly distributed** across Greater Chennai
- **Central zones** show consistently high accessibility due to better service concentration
- **Peripheral zones** (Manali, Thiruvotriyur, Ambattur, Madhavaram) show **low accessibility levels**
- Hospitals contribute the most to accessibility due to **wider service coverage**
- Parks are **limited and uneven**, indicating gaps in recreational infrastructure

PLANNING IMPLICATIONS

- Need to **improve service distribution** in low-access zones
- Prioritize development of **healthcare and educational facilities** in peripheral areas
- Increase **urban green spaces (parks)** for balanced liveability
- Use GIS-based analysis for **data-driven planning decisions**
- Support **equitable infrastructure development** across all zones

GitHub link: <https://github.com/kondojukushi10-rgb/chennai-service-accessibility-analyzer>